Can Satellites Replace the Field Plot? A Blind Benchmark of Satellite Forest Carbon and Conformal Crediting with Calibrated Risk

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***Abstract*—Satellite platforms promise to replace field inventories in forest carbon crediting, but their estimates are rarely tested blind against the standard-formula answer, and credits are issued with deductions of unknown risk. We separate the question into two isolated grids. The right grid computes sealed field ground truth to SNI 7724, IPCC, CDM, and Verra rules. The left grid sees only plot coordinates and dates and estimates carbon from time-matched Sentinel-1/2, Landsat, PALSAR, canopy height, and GEDI lidar. On 476 plots, blind error was 21% on 1 ha plots and 7–9% on project totals outside Indonesia, but near 100% on small Indonesian plots. Counting trees on a 1 m canopy height map and applying the field formula found 3% of the stems and 25% of the carbon. In 1,000 simulated projects the best map found a median 95% of the field carbon. Only 21.5% of Sentinel-2 acquisitions see an Indonesian plot clear, and a common scene filter discards 65% of those. We propose conformal crediting, a lower bound calibrated on other projects with field truth. It met a 10% over-crediting target for five maps within 2.9 points, was invariant to map inflation, and credited 72–84% of the true stock without a field plot, while the VM0047 deduction over-credited 23–29% of projects that had field plots. No satellite estimate of annual growth beat assuming no change.**

***Index Terms*—forest carbon, MRV, remote sensing, cloud cover, conformal prediction, spatial cross-validation, carbon crediting, commit-reveal ledger, distributed processing**

I. Introduction

A forest carbon credit is issued on a number computed by a fixed chain. Field crews measure each stem's diameter D and height H and identify its species for wood density ρ; a pantropical allometric equation converts these to aboveground biomass (AGB) [1], and fixed factors convert AGB to carbon and CO_2 [2], [3]. The chain is the standard. What is costly is the field visit, above all in remote Indonesian forests. Satellite platforms, including our own NusaCarbon prototype, promise to replace the visit, and global biomass maps exist [4], [5]. Investigations of REDD+ crediting, however, show that unverifiable estimates produce over-crediting [6].

How close is a satellite number to what the standard formula gives at the same place and time? Three traps inflate reported accuracy: *leakage*, when a model is trained or tuned on plots next to those it is scored on [7]; *temporal mismatch* between the field and image dates; and *scale*, when 0.05 ha plots are compared with 10 m to 4.6 km pixels. An audit of our own platform found forest figures driven by default constants, scene-level cloud screening, and a metadata "confidence" grade, none checked against field data.

We separate the problem into two grids (Fig. 1). The **right grid** is fixed: public field plots, the standard formula, a carbon density. The **left grid** is free, any satellite data and any algorithm, subject to one rule: it never sees the right grid's outcomes for the plots it is scored on. We ask: **RQ1** (methodology) how far a blind satellite estimate deviates from standard-formula ground truth, by plot size and aggregation; **RQ2** (AI) which inputs and learners close the gap, and whether trees can be counted from space; **RQ3** how often clouds hide an Indonesian plot, and whether uncertainty can be stated honestly at plot and project scale; **RQ4** whether a credit can be issued at a stated, verified risk of over-crediting, and whether growth can be estimated; and **RQ5** (distributed systems) whether extraction can be sharded and made fault tolerant with bit-identical output.

Contributions. (i) A blind, time-matched protocol with a commit-reveal ledger (476 stock and 196 repeat-census plots). (ii) A standards-based right grid with a Monte Carlo uncertainty budget. (iii) A plot-level clear-sky climatology for Indonesia and the error rate of the scene cloud filter. (iv) A test of missing-data conformal intervals under cloud. (v) *Conformal crediting*, a crediting rule with a calibrated over-

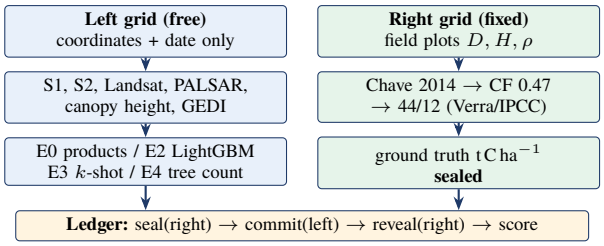


Fig. 1. Blind two-grid design. Only the manifest (plot id, coordinates, date, radius) crosses from right to left.

crediting risk that is invariant to map inflation. (vi) A tree-by-tree satellite estimator benchmarked against field stem lists, and 1,000 project simulations reported as carbon amounts. (vii) A sharded, idempotent extraction runner. Conformal crediting reuses standard conformal machinery [8] for a new decision.

II. Related Work

Remote sensing. Optical and radar signals saturate over dense canopy. GEDI lidar samples canopy height in 25 m footprints [9]; global maps include ESA CCI Biomass [4], Chloris [5], and 1 m and 10 m canopy height [10], [11]. Sentinel-1 and PALSAR reached 31–38% in Kalimantan against a lidar layer [12], and airborne lidar 13% at 1 ha in Sabah [13].

Validation and uncertainty. The BIOMASS package propagates the uncertainty of Chave's allometry [14]. Conformal prediction gives finite-sample intervals [8]; conformalized quantile regression (CQR) adapts their width [15], and Zaffran *et al.* extend conformal prediction to missing covariates [16]. Conservative crediting asks that a credited stock exceed the truth with small probability [17], which Neeff solves parametrically [18].

III. Methodology

A. Right Grid: Ground Truth by the Standards

Data. The stock benchmark has 476 plots in 27 region blocks: 148 HCSA plots in six Indonesian provinces, nested 0.01/0.05 ha [19]; 142 GEO-TREES network plots, pantropical [20]; and 186 Labrière 1 ha units in French Guiana and Gabon [21]. For change, the SAFE census in Sabah gives 25×25 m plots measured stem by stem with day-precision dates from 2010 to 2020 [22]. We follow SNI 7724 [23] (five pools, precision ≤20%), the HCSA toolkit [24], IPCC [3] (Monte Carlo allowed), CDM AR-TOOL14 [25], Verra VM0047 [2] (deduct $U - 10\%$), and Gold Standard [26] (deduct the excess over 20%).

What a plot is. Coordinates only locate a plot; the ground truth is the list of trees measured inside it. A crew lays out the plot around its centre, tags every stem above the minimum diameter, measures its diameter D at 1.3 m with a tape and its height H with a clinometer, and records its local name, which a botanist resolves to species and family. Wood density ρ, the oven-dry mass per green volume, is then looked up by species, or the genus or family mean. HCSA publishes the list: 2,528 stems of 370 species in 89 families, each with D, H, name, and the ρ of the ICRAF database [19]. GEO-TREES and Labrière publish plot biomass computed from their own lists; none of it reaches the left grid.

Tree, plot, and carbon. A tree's AGB follows Chave 2014 Eq. 4 [1],

$$\mathrm{AGB}_i = 0.0673\,(\rho_i D_i^2 H_i)^{0.976}\ [\mathrm{kg}], \qquad \sigma_{\ln} = 0.357, \quad (1)$$

with H measured (HCSA) or from a local $\ln H$–$\ln D$ model (Labrière) [14]. Tree biomass is expanded by the area in which a stem of its size is counted, and pools and carbon follow IPCC and SNI 7724:

$$\mathrm{AGB}_j = \sum_{i \in j} \frac{\mathrm{AGB}_i}{1000\, a(D_i)}, \quad a(D) = \begin{cases} 0.01, & 5 \le D < 15 \\ 0.05, & D \ge 15 \end{cases}$$
$$C_j = \mathrm{CF}\,(1+R)\,\mathrm{AGB}_j, \qquad \mathrm{CO_2e}_j = \tfrac{44}{12}\,C_j, \quad (2)$$

with CF = 0.47 and root-to-shoot $R = 0.37$ (SNI 7724, IPCC 2006) or 0.21 (IPCC 2019). The benchmark compares AGB, so CF and R cancel and every relative error holds for t C ha^{-1} and t CO$_2$e ha^{-1}. HCSA was recomputed tree by tree; plots with a missing input or an implausible stem (a *Gluta renghas* at $D = 480$ cm, a circumference typed as a diameter) were excluded, never imputed, and palms because Eq. 1 is for dicotyledonous trees.

Uncertainty and deduction. Plots are grouped into strata h with weights w_h, and the project uncertainty at 90% is $u = t_{0.95}\sqrt{\sum_h w_h^2 s_h^2/n_h}/\bar{b}$ (AR-TOOL14) [25]. AR-TOOL14 counts sampling error only, so we add IPCC Approach 2: 2,000 Monte Carlo draws per tree of diameter, height, wood density, allometric residual, and CF errors with the BIOMASS error models [14]. For change on the same plots, VM0047 v1.1 Eq. 28 gives $U = T\sqrt{SE_0^2 + SE_1^2 - 2\rho\, SE_0 SE_1}/\Delta C$, where ρ is the correlation of the two stock means; credits are multiplied by $1 - \min(1, \max(0, U - 0.1))$ [2].

B. Left Grid: Inputs and Estimators

For each plot, using only coordinates and the field date t_j, the left grid reads clear observations within $t_j \pm 90$ days: Sentinel-2 L2A bands and indices, Landsat 5–9, Sentinel-1 backscatter, the annual ALOS PALSAR mosaic, and elevation. Through Google Earth Engine it also reads the Meta 1 m and ETH 10 m canopy height maps [10], [11], GEDI L4A biomass and L2A height within 1 km [9], ESA CCI Biomass [4], and Hansen forest loss [27]. **E0** reads published products as they are. **E2** is LightGBM [28] on ln AGB, trained on all region blocks except the target and predicting it blind (leave-one-region-out, LORO), with smearing retransformation. **E3** recalibrates E2 by the ratio on k random plots of the target block. We score relative RMSE (rRMSE), bias, R^2, and the error of the mean of n plots, which approximates a project total, with a region-block bootstrap (2,000 draws) and five seeds.

From pixels to carbon. A satellite does not see trees; it records reflected light and radar echoes per pixel. Optical

bands become indices such as the normalized difference vegetation index [29]

$$\mathrm{NDVI} = (R_{\mathrm{NIR}} - R_{\mathrm{red}})/(R_{\mathrm{NIR}} + R_{\mathrm{red}}), \tag{3}$$

with R the reflectance in Sentinel-2 bands B8 (842 nm) and B4 (665 nm): chlorophyll absorbs red light and leaves reflect near-infrared, so a green canopy reads about 0.8 and bare soil about 0.1–0.2. The estimator then has four steps: (1) read every clear pixel of the plot window within ± 90 days; (2) summarise each input into features (band medians, NDVI and six other indices, backscatter, canopy height percentiles, GEDI biomass); (3) predict ln AGB with a model trained on the plots of other regions; (4) convert with the right grid's factors, $C = 0.47\,\mathrm{AGB}$ and $\mathrm{CO_2e} = \frac{44}{12}C$.

E4: count the trees, then the formula. E4 rebuilds the right grid's tree list from the 1 m canopy height map, smoothed by a 1 m Gaussian. Tree tops are local maxima above 5 m in a window of radius $0.06H + 0.5$ m [30]; a crown grows from each top while the canopy stays above half its height; and the stem diameter follows from height and crown diameter CD by the tropical-angiosperm model of Jucker *et al.* [31],

$$D = a\,(H \cdot CD)^b \text{ [cm]}, \qquad (a, b) = (0.486,\, 0.802), \tag{4}$$

for Indo-Malaya (0.428, 0.821 Afrotropic; 0.591, 0.781 Neotropic). A satellite cannot identify species, so every tree takes the mean ρ of its region's tropical species in the Global Wood Density Database (0.574 for Southeast Asia) [14], and Eq. 1 gives its biomass. E4 uses no field data; *E4 scaled* multiplies it by the field-to-E4 ratio of the other regions.

C. Clouds

For 97 locations we read the Sentinel-2 scene classification of all 22,697 acquisitions of 2021–2023 in a 5×5 window of 20 m pixels; an acquisition is *clear* if every pixel is. We test the common scene filter (`CLOUDY_PIXEL_PERCENTAGE` < 30) against this condition. For a share q of target plots we then remove the optical block and compare a naive model, modality dropout [32], a switch to a radar model, and imputing optical features from radar.

D. Confidence: Plot and Project Intervals

All intervals target 90% and are calibrated on inner LORO residuals of the training regions only. CQR [15] scores $s = \max\{\hat{q}_{\mathrm{lo}}(x) - y,\, y - \hat{q}_{\mathrm{hi}}(x)\}$ on $y = \ln \mathrm{AGB}$ and widens the quantile band by the conformal quantile of s. *Cloud-aware* CQR scores each calibration plot twice, as observed and with its optical block removed, and gives each test plot the threshold of its own missingness pattern, a two-pattern case of CP-MDA [16]. Its guarantee assumes missingness independent of the outcome, which clouds violate, so coverage must be measured. A project's satellite mean has errors that share one regional bias, so its standard error is $SE^2 = s_w^2/n + s_b^2$, with s_w and s_b the within- and between-region spread of log residuals in other regions. With k field plots, the satellite becomes auxiliary data for a ratio estimator.

E. Conformal Crediting

A registry issues a number, not an interval. It should be a lower bound that exceeds the true stock in at most a share δ of projects [17], [18]. For a satellite map no design-based standard error exists, so we calibrate the bound on other projects with field truth. Project j has estimate $\hat{Y}_j$, true mean Y_j, and relative standard error r_j ($r_j = 1$ if none); its score is $z_j = \ln(\hat{Y}_j/Y_j)/r_j$. With scores from m calibration projects, a new project is credited

$$C = \hat{Y}\exp(-q\,r), \qquad q = z_{(\lceil (m+1)(1-\delta) \rceil)}. \tag{5}$$

If projects are exchangeable, $P(C > Y) \le \delta$, as in split conformal prediction with projects as units [8]; plots inside a project may be correlated in any way. The rule needs $m \ge 1/\delta - 1$ projects, nine for $\delta = 0.1$. If every map value is multiplied by c, every plain score shifts by $\ln c$, so q shifts by $\ln c$ and C is unchanged: *inflating a map earns no credit*. We split regions of 16 or more plots into spatial sub-projects of about 10 plots, giving 31 projects in 19 regions, and calibrate each project only on projects from other regions. We score the realised over-crediting rate and the *yield*, the mean share C_j/Y_j of the true stock credited.

F. Ledger and Distributed Extraction

Every protocol step is an entry in an append-only log carrying the SHA-256 of the previous entry [33]: `seal` (digest of the sealed outcomes), `commit` (digest of each prediction file), `release` (training labels per fold), and `reveal`, accepted only if the sealed file still matches; commits after a reveal are refused. An extraction task is one (plot, date, window, sensor) request stored under its hash, so re-execution is idempotent and the cache is a checkpoint. Transient errors are retried with capped backoff, and dense tiles are split and placed by consistent hashing [34] or the longest-processing-time (LPT) rule [35].

IV. Results

A. How Certain Is the Right Grid?

On the 148 HCSA plots, the Monte Carlo 95% interval of plot AGB has a median half-width of 31%. The standards' own choices are large: the SNI 7724 root-to-shoot ratio gives 13% more carbon than the 2019 Refinement, and the same Jambi data receive deductions of 24% (VM0047), 34% (AR-TOOL14), and 14% (Gold Standard). AR-TOOL14 counts sampling uncertainty only (9–34% by province), though measurement and allometric error carry 10–64% of the variance of the mean. For growth, VM0047 Eq. 28 with $\rho = 0$ deducts all credit for 13 of 15 SAFE blocks; with the measured ρ (0.45–0.99), seven keep some.

B. Blind Accuracy (RQ1) and Closing the Gap (RQ2)

Table I gives the blind errors. The confirmatory E2, committed before the reveal, missed by 48.5% overall, 28.6% on the 1 ha units, 41.9% on GEO-TREES, and 102.9% on HCSA, where neither committed estimator beat predicting the mean. Plot size dominates the error. The largest gain came from

TABLE I
BLIND RRMSE (%) AGAINST THE RIGHT GRID, LEAVE-ONE-REGION-OUT

Left-grid estimate	All	GEO-T.	HCSA	1 ha
E0 Chloris, as published	49.2	37.4	107.1	30.7
E0 ESA CCI, as published	49.4	38.8	105.4	31.5
E0 GEDI L4A, as published	59.2	50.0	107.0	46.6
E2 (S1/S2, Landsat, PALSAR, DEM)	49.1	42.0	104.8	28.8
+ GEDI	45.5	36.6	102.4	24.5
+ canopy height	41.5	32.2	**94.8**	22.4
+ canopy height + GEDI	**40.1**	**27.7**	95.6	**20.6**
Gain over E2 (points)	−9.0	−14.3	−9.2	−8.2

475 plots with AGB $>1\,\mathrm{Mg\,ha^{-1}}$: GEO-TREES 142, HCSA 148 (0.05 ha), Labrière 185 (1 ha). Region-bootstrap 95% intervals of the gain row all exclude zero.

inputs, not algorithms: adding the 1 m canopy height map and GEDI cut the overall error by 9.0 points (bootstrap 95% interval -13.0 to -6.1), to 20.6% on the 1 ha units and 27.7% on GEO-TREES, and raised R^2 from 0.29 to 0.53. Read as published, GEDI under-estimated by 29–41%: lidar helps as an input, not as an answer.

Where the error comes from. Predictions saturate: above $400\,\mathrm{Mg\,ha^{-1}}$ (mean 480) the E2 mean is 293, and below 50 (mean 21) it is 60, while between 100 and $300\,\mathrm{Mg\,ha^{-1}}$ the median error is only $+5$ to $+12$%. The HCSA plots hold a median of 15 stems, so one large tree moves a plot by hundreds of $\mathrm{Mg\,ha^{-1}}$, which no 10–30 m pixel can resolve.

Aggregation and local plots. With canopy height and GEDI and no field plot, the median error of a block total was 7% on the 1 ha units, 9% on GEO-TREES, and 20% on HCSA, and the mean of 20 plots was within 3–5% of its field mean outside Indonesia. A ratio correction on $k = 10$–20 local plots (E3) brought E2 block totals from 29% to 9–10% on GEO-TREES and from 11% to 6% on Labrière, but hurt on HCSA until $k = 20$.

C. Counting Trees from Space, and Ten Projects

What the field counts. The median HCSA plot holds 700 stems per hectare of five species. Most stems are small and most carbon sits in a few large trees: stems of 5–15 cm are 75% of the count but 13% of the carbon, and the 6% of stems of 30 cm or more hold 58%. By wood density, light wood ($\rho < 0.5\,\mathrm{g\,cm^{-3}}$, mostly Lauraceae and Moraceae) is 24% of the stems and 16% of the carbon, medium wood (0.5–0.7, led by Dipterocarpaceae) 44% and 44%, and heavy wood (≥ 0.7) 32% and 40%. Giving every stem the regional default ρ instead of its species value leaves 91% of the total carbon, and 78–115% of a plot's (10th to 90th percentile).

What the satellite counts. On the HCSA plots E4 found a median of 20 trees per hectare: 3% of the stems, 11% of those of 15 cm or more, with a rank correlation of 0.18 against the field count. The map stores whole metres and is smoother than a crown: without smoothing every stair step becomes a top (270 per hectare), so the count is a processing choice, not a measurement. E4 found 25% of the field carbon, 23–26% in every dataset. Scaled on other regions ($\times 3.8$), it found 97% of the total, but only 36% of plots fell within ± 25% of the field value, against 60% for the map. Indices alone cannot do better: of the 136 plots with a clear Sentinel-2 view, the median NDVI (Eq. 3) was 0.77 below $100\,\mathrm{Mg\,ha^{-1}}$ and 0.83 above 300, and above $150\,\mathrm{Mg\,ha^{-1}}$ its rank correlation with biomass was -0.13. A closed canopy is equally green at any biomass.

TABLE II
TEN SIMULATED 1,000 HA PROJECTS OF 10 RANDOM PLOTS: CARBON FOUND BY EACH GRID, % OF THE FIELD AMOUNT

#	Region	Field, kt	Map	E4	E4 sc.	Credit
1	HCSA Riau	238	71	21	80	64
2	Labrière Nouragues	686	99	23	88	90
3	Labrière Mondah	338	84	16	60	77
4	Labrière Mabounie	594	87	19	71	79
5	HCSA West Papua	254	91	46	180	83
6	HCSA Riau	191	96	8	32	87
7	Labrière Rabi	472	107	33	126	97
8	Labrière Mabounie	556	92	19	72	83
9	HCSA Jambi	287	93	23	88	86
10	Labrière Mondah	156	112	12	45	102
1,000 projects: median			95	23	85	86
5th–95th percentile			46–115	4–44	17–168	42–106
within ±25% of the field			80	0	40	76

Field: $\mathrm{kt\,CO_2e}$ from the measured trees, Eq. 1, $\times 0.47 \times \frac{44}{12}$. Map: E2 + canopy height + GEDI. E4 sc.: scaled on other regions. Credit: conformal, $\delta = 0.1$. HCSA projects: field 570–924 $\mathrm{stems\,ha^{-1}}$, E4 14–18.

Ten projects. Table II reports, without RMSE, how much carbon each grid finds in a simulated 1,000 ha project of 10 random plots from one region: seeds 1–10 of 1,000, with per-plot rows in the extended version. Over all 1,000, the map found a median 95% of the field carbon, within ± 25% in 80% of projects. Conformal crediting issued a median 86% and exceeded the field amount in 12% of these random projects, near its 10% target.

D. Clouds and Confidence (RQ3)

How often is a plot clear? At a median Indonesian location only 21.5% of Sentinel-2 acquisitions see the plot clear (7.6% in January, 33.5% in May; the Americas 39%), but revisit compensates: the median gap between clear looks is 10 days. *The scene filter is the real problem*: in Indonesia it keeps 11.5% of acquisitions, 34% of them cloudy over the plot, and discards 65% of the clear ones. Masking at the plot gives 2.8 times as many clear looks. Removing every optical feature raised the error only from 48.0% to 50.0%, because the model leans on radar, terrain, and annual products; imputing optical features from radar was harmful (bias -61%).

Which intervals are honest? CQR is the only interval method that reaches the 90% target with usable width (89.9%, upper bound 5.3 times the lower). Split conformal covers 88.4% at a ratio of 16.3, and normalized conformal under-covers (79.4%) because a difficulty model fitted on other regions does not transfer. When every target plot is clouded, plain CQR falls to 86.5%, and cloud-aware CQR keeps 90.5% by widening honestly ($+4.0$ points, interval $+1.9$ to $+6.6$). HCSA stays under-covered (68–81%) by every method. A metadata confidence score like our platform's carries no infor-

TABLE III
CREDITING RULES AT $\delta = 0.10$, 31 PROJECTS, CROSS-REGION CALIBRATION

Rule	Over-credited	Yield
Satellite only (no plot in the project)		
No deduction	35.5%	94.4%
Parametric bound, independent errors	9.7%	72.6%
Conformal (Eq. 5)	9.7%	72.2%
Conformal, canopy height + GEDI map	9.7%	84%
Satellite + 5 random field plots (ratio estimator)		
VM0047 ($U - 10\%$)	22.6%	82.2%
Design lower bound	9.9%	80.7%
Conformal, studentized	9.6%	80.7%

mation about error once the dataset is controlled for (partial rank correlation -0.03 to $+0.07$).

Projects. For 19 projects (blocks of eight or more plots), a naive interval with independent plot errors covered only 74% (HCSA 33%); the two-component interval covered all 19, at a half-width of 71–86%. With field plots, the satellite as auxiliary data cut the plots needed for $\pm 10\%$ from 408 to 216 (GEO-TREES) and from 381 to 162 (Labrière), and none on HCSA.

E. From Intervals to Credits and Growth (RQ4)

Table III credits the 31 projects at $\delta = 0.10$. Crediting the raw map over-credits 35% of projects. VM0047 on the honest two-component uncertainty never over-credits but credits only 48% of the true stock. Conformal crediting meets its target: 9.7% of projects are over-credited, and 72% of the true stock is credited without a field plot in the project, 84% with the canopy height and GEDI map. A better map buys credits; it does not buy risk. With field plots, the VM0047 deduction still over-credits 23% of projects with 5 plots and 29% with 10 (yield 90%), because the deduction only starts above 10% uncertainty. The design lower bound at the same δ meets the target (8.5% with 10 plots, yield 86%).

Whatever the map? Applied to five maps (E2, Chloris, their average, and E2 inflated by 10% and 20%), the conformal bound's realised over-crediting stayed within 2.9 points of the target, against 9.4 for the parametric bound. A map inflated by 20% earns 87% instead of 73% and doubles the over-crediting rate under the parametric bound, and the same 72.2% under conformal crediting. Varying δ traces an integrity-yield frontier that prices a field plot: at about 10% risk the satellite alone earns 72% of the true stock, 5 plots 81%, 10 plots 86%, and both VM0047 points lie below it. With a bank of six to eight projects of one type, yield fell to 18%.

Growth. Excluding oil palm (Chave's equation gave $505\,\mathrm{Mg\,ha^{-1}}$), 196 SAFE plots gained a mean $1.87\,\mathrm{Mg\,ha^{-1}\,yr^{-1}}$. At plot level, the Chloris difference had the same error as predicting no change (RMSE 8.06 vs 8.12), and the E2 stock difference was twice as bad with the wrong sign: a stock error of 30–100% cannot resolve growth of 1% per year. The mean rate of the other blocks was the most accurate estimator (block MAE 0.98 vs 2.05 for no change), and PALSAR, CCI change, and Hansen loss did not beat it (1.20).

F. Systems (RQ5)

The stock extraction ran 2,052 tasks on 24 workers in 2.1 h. Splitting dense tiles cut the worst-worker load factor from 6.75 to 1.40 (LPT), and retries absorbed 85 transient errors. Results were bit-identical to the single-worker run at every worker count, under injected faults, and after a resume, and the 37-entry ledger verifies end to end.

V. DISCUSSION

Can satellites replace the field plot? Not at plot scale, and not for growth. Tested blind, open satellite data put a 1 ha plot's stock within about 21%, against 13% for airborne lidar in Sabah [13] and 31–38% reported for radar in Kalimantan against a lidar layer [12], but a 0.05 ha plot within about 100%. Nor can they count trees: the 1 m map shows a canopy surface, not stems, and the field formula applied to what it shows finds a quarter of the carbon. For growth, the quantity credits are paid on, no satellite estimator beat a regional average of measured growth. What satellites do well is *replace part of the inventory*: project totals within 7–9% outside Indonesia, and half the field plots where plots are large enough to rank.

The ground truth is a model too. Two valid standards differ by 13% in total carbon, and deductions for the same data range from 14 to 34%. "100% accuracy" has no meaning; the target is the fully computed uncertainty of a field estimate made to a stated standard.

Confidence must be calibrated, not graded. A metadata score said nothing about error, and the $U - 10\%$ deduction is not a lower confidence bound. Conformal crediting makes the over-crediting rate a design parameter and removes the reward for inflating the map. Its novelty is narrow: the target is Grassi's and Neeff's, the machinery is split conformal prediction, and conformal bounds for area-level parameters exist [36]; we found no crediting rule calibrated across projects against field truth and tested under transfer and biased maps.

Implications and roadmap. For NusaCarbon the evidence implies per-pixel cloud masking at the plot, a radar-led model with canopy height and GEDI, a cloud-aware CQR interval instead of a letter grade, conformal crediting from a bank of at least nine field-verified projects, and growth from remeasured permanent plots. Canopy structure gave the largest gain and stems are what the satellite cannot see, which motivates the planned drone tiers: a DJI Matrice 350 RTK with a Zenmuse L2 lidar above the canopy, and an under-canopy swarm for stem diameters, where piloted scanning has reached 0.6–0.9 cm DBH error [37]. Three hypotheses are to be tested on Indonesian 1 ha plots: swarm DBH within 3% of tape, lidar AGB within 15%, and repeat-flight growth beating the regional default rate.

Threats to validity. Only E0, E2, and the first change estimates are confirmatory; all later experiments, E4 included, were defined after the reveal and are exploratory, and we report every variant we ran. GEDI and Chloris were calibrated on

field data that may include some of our sites. HCSA and Labrière dates are known only to the year. Projects are region blocks, not registered projects.

VI. Conclusion

We tested blind, against a sealed standard-formula right grid, whether open satellite data can replace the field plot. Satellites reached 21% error on 1 ha plots, 7–9% on project totals, and a median 95% of the field carbon in 1,000 simulated projects, but failed on small plots, tree counts, and growth. Plot-level cloud masking nearly triples usable optical data, cloud-aware CQR keeps intervals honest, and conformal crediting issues credits at a stated risk without a field plot, unaffected by map inflation.

Data and Code Availability

Field data are public [19]–[22]. Code, results, an extended version with all tables, and an interactive dashboard are at https://github.com/kennytangg/carbon-paper.

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